



EUROPEAN
COMMISSION

Brussels, 8.6.2023
COM(2023) 304 final

**REPORT FROM THE COMMISSION TO THE EUROPEAN PARLIAMENT, THE
COUNCIL, THE EUROPEAN ECONOMIC AND SOCIAL COMMITTEE AND THE
COMMITTEE OF THE REGIONS**

**identifying Member States at risk of not meeting the 2025 preparing for re-use and
recycling target for municipal waste, the 2025 recycling target for packaging waste and
the 2035 municipal waste landfilling reduction target**

{SWD(2023) 175 final} - {SWD(2023) 176 final} - {SWD(2023) 180 final} -
{SWD(2023) 181 final} - {SWD(2023) 182 final} - {SWD(2023) 183 final} -
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{SWD(2023) 198 final} - {SWD(2023) 199 final} - {SWD(2023) 200 final}

1. INTRODUCTION

Turning waste into resources is an essential building block of the circular economy. Proper implementation of the EU's waste legislation involves the application of the waste hierarchy, which gives priority to preparing for re-use and recycling and puts landfilling as the least preferred waste treatment option. The EU's targets for the treatment of municipal and packaging waste have been established in order to ensure concerted efforts in all Member States to move up the waste hierarchy.

Improving performance in waste management will speed up the transition to a circular economy and contribute to the objectives of the circular economy action plan for a cleaner and more competitive Europe¹ to double the EU circular material use rate, significantly reduce total waste generation and halve the amount of residual municipal waste by 2030. Furthermore, sound waste management is essential to prevent the negative impact of waste generation on the environment and health, and to achieve the targets of the zero pollution action plan² on waste and marine litter.

Increasing preparing for re-use and recycling and reducing landfilling are key to reaching climate neutrality, increasing the security of supply of raw materials, saving energy, and reducing EU dependencies on imports from third countries, while creating local jobs and boosting innovation in new technologies for sustainable products and materials management.

To better reflect the EU's ambition to move to a circular economy, in 2018 several amendments to EU waste directives were adopted, including ambitious targets for the preparing for re-use, recycling and landfilling of municipal and for the recycling of packaging waste. Municipal waste represents around 10% of the total waste generated in the EU³, but it is one of the most complex streams to manage due to its diverse composition, the large number of waste producers and the fragmentation of responsibilities for its management. A large share of municipal waste is composed of packaging waste, which has significant circularity potential.

In order to ensure better, more timely and uniform implementation of these ambitious targets and to anticipate any implementation weaknesses, the EU rules adopted in 2018 established a system of early warning reports to detect shortcomings and allow action to be taken ahead of the deadlines for meeting the targets. Under these rules, the Commission is tasked, in cooperation with the European Environment Agency, to draw up reports on the progress of Member States towards the attainment of the targets.

This overall report assesses Member States' likelihood of meeting the 2025 targets set out in Article 11(2)(c) of Directive 2008/98/EC on waste (the Waste Framework Directive)⁴ and in Article 6(1)(f) of Directive 94/62/EC on packaging and packaging waste (the Packaging Directive⁵). The targets require Member States to take measures to achieve a minimum of:

- 55% preparing for re-use and recycling of municipal waste,
- 65% recycling of all packaging waste, and

¹ COM(2020) 98 final.

² COM(2021) 400 final.

³ Eurostat

⁴ <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A02008L0098-20180705>

⁵ <https://eur-lex.europa.eu/legal-content/en/ALL/?uri=celex%3A31994L0062>

- material-specific packaging waste recycling targets of 75% for paper and cardboard, 70% for glass, 50% for aluminium, 50% for plastic and 25% for wood.

The report also provides a preliminary assessment of the likelihood of Member States meeting the target laid down in Article 5(5) of Directive 1999/31/EC on the landfill of waste (the Landfill Directive) to reduce landfilling of municipal waste⁶ to less than 10% by 2035. The report summarises main findings, key recommendations to Member States at risk of not achieving certain waste targets, and examples of best practices of waste management intended to help improve waste recycling performance. It is complemented by staff working documents containing country-specific information and recommendations for those Member States at risk of missing the 2025 preparing for re-use and recycling targets for municipal waste and all packaging waste, which include the following:

- an estimation of the attainment of the targets by each Member State;
- a list of Member States at risk of not attaining the targets within the respective deadlines, accompanied by appropriate recommendations for the Member States concerned; and
- examples of best practices that are used throughout the EU and could provide guidance for making progress towards the targets.

The work carried out for the purposes of this early warning report builds on continuous support and cooperation between the Commission and Member States on improving the implementation of EU waste legislation. Following compliance promotion exercises starting in 2013⁷, in 2018 the European Commission issued a set of reports on progress and implementation challenges for several waste streams⁸ that included possible actions to support Member States' efforts to improve their waste management performance⁹. In addition, since 2017 the European Commission has been regularly publishing an Environmental Implementation Review (EIR), a tool designed to improve Member States' implementation of EU environmental laws and policies. The latest EIR was issued in 2022¹⁰.

2. METHODOLOGY

This report is the result of an extensive consultation and collaboration among the European Commission, the European Environment Agency, Eurostat and the Member States. The assessment was based on a methodology developed by the European Environment Agency, which took as a starting point the waste management data submitted by the Member States to Eurostat in line with reporting requirements under the waste directives, and considered existing and planned policies at national level and other relevant information. It also considered the adoption of revised national waste prevention programmes as required by the Waste Framework Directive.

This assessment allowed the European Environment Agency¹¹ to identify the Member States likely to miss the targets. Its findings were further discussed between the European

⁶ <https://eur-lex.europa.eu/legal-content/en/TXT/?uri=CELEX%3A31999L0031>

⁷ [Implementation of the Waste Framework Directive \(europa.eu\)](#)

⁸ COM(2018) 656 final.

⁹ https://environment.ec.europa.eu/topics/waste-and-recycling/implementation-waste-framework-directive_en

¹⁰ https://environment.ec.europa.eu/law-and-governance/environmental-implementation-review_en#country-reports

¹¹ <https://www.eea.europa.eu/publications/many-eu-member-states-are>

Commission and the Member State concerned; this enabled additional information to be gathered and led to the finalisation of the report. A call for evidence was used to inform about the initiative and gather stakeholder views¹².

This report is based on the latest Eurostat data available at the stage of its preparation. The data for reference year 2020 was used for municipal waste, while for packaging waste the data for 2019 was used. Regarding packaging, more recent data for 2020 is now available and a preliminary assessment of its relevance to the conclusions of the report is reflected in Section 5.

3. OVERALL FINDINGS

The analysis of Member States' performance identified the general trends in waste management and showed the likelihood of achieving waste preparing for re-use and recycling targets by 2025.

The majority of Member States have recently carried out or are in the process of implementing waste reforms that will result in increased recycling rates in the years leading to 2025 and beyond. The implementation of the 2018 waste package and the recommendations issued in the 2018 Early Warning Report played a relevant role in achieving progress. However, in some cases delays in their application are affecting national recycling performances.

There are still significant differences in waste management performance across Member States. For some countries, the distance from the targets is very large and greater efforts will be needed to implement reforms on the ground (e.g. as regards biowaste treatment, separate collection, data quality, etc.). There is also evidence that external factors influenced performance, for instance during the COVID pandemic separate collection in some Member States was reduced or halted. There are also indications that the recent spike in energy prices is adversely affecting recycling activities.

Overall, according to the assessment carried out by the European Environment Agency¹³, **the majority of Member States are at risk of missing the municipal waste preparing for re-use and recycling targets for 2025.**

More specifically, with regard to the target of 55% preparing for re-use and recycling of municipal waste and 65% recycling of all packaging waste to be achieved by 2025:

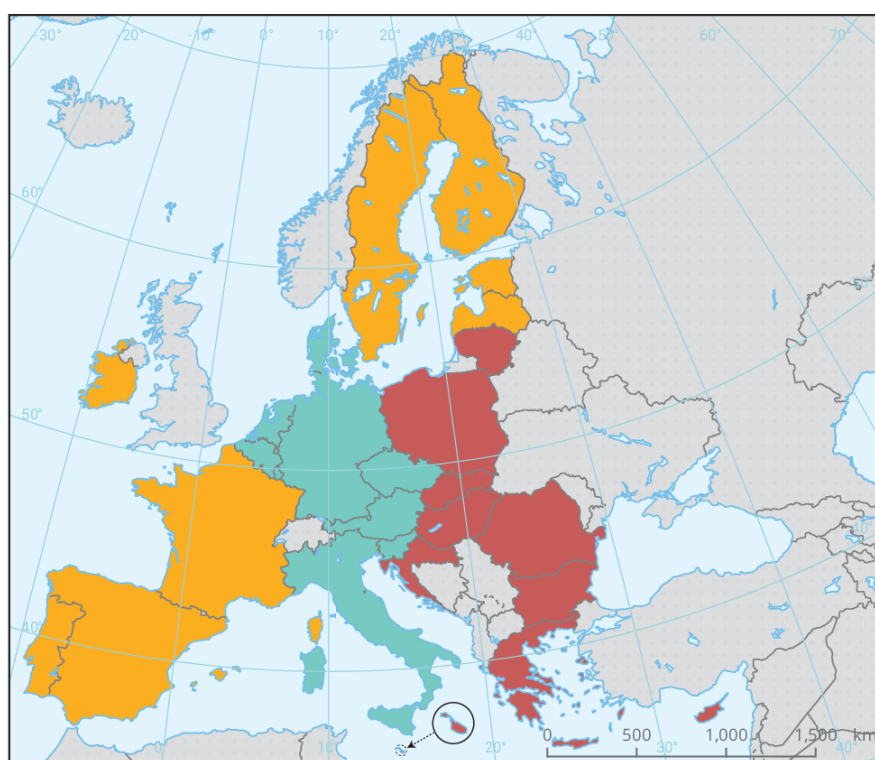
- **9 Member States are on track to meet both targets** – Austria, Belgium, Czechia, Denmark, Germany, Italy, Luxembourg, the Netherlands and Slovenia;
- **8 Member States are at risk of missing the municipal waste target only**, but not at risk for the target on all packaging waste – Estonia, Finland, France, Ireland, Latvia, Portugal, Spain and Sweden;
- **10 Member States are at risk of missing both the targets for municipal and all packaging waste for 2025** – Bulgaria, Croatia, Cyprus, Greece, Hungary, Lithuania, Malta, Poland, Romania and Slovakia.

¹²https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/13511-Early-Warning-Report-on-Waste_en

¹³<https://www.eea.europa.eu/publications/many-eu-member-states-are>

This early warning report is accompanied by staff working documents for Member States considered at risk of missing at least one of these two headline targets (55% preparing for re-use and recycling target for municipal waste and 65% recycling target for all packaging waste to be achieved by 2025); this corresponds to the second and third groups of Member States listed above¹⁴. All Member States at risk of missing the target for all packaging waste are also at risk of missing the municipal waste target.

It should be noted that achieving high performance in re-use, preparing for re-use and recycling of waste requires continuous efforts both in Member States that have been identified at risk of missing 2025 targets and in those for which the assessment has shown likelihood to achieve them although they still have some distance to cover. As ensuring proper waste management in line with the waste hierarchy often faces new challenges, also those countries that have reached preparing for re-use and recycling rates above the 2025 targets in 2019 and 2020 need to pursue policies to maintain and further improve performance levels, including in the context of the application of the new calculation rules for recycling. Further efforts should be made to achieve material specific recycling targets for packaging, in particular for plastic packaging.



Reference data: ©ESRI

- Member states not at risk for both targets
- Member States at risk for the municipal waste preparing for re-use and recycling target but not at risk for the all packaging waste recycling target
- Member states at risk for both targets
- Outside coverage

¹⁴ The assessment of the risks of missing the targets takes into account data and information available until April 2022.

Figure 1: Prospects for EU Member States of meeting the preparing for re-use and recycling targets for municipal waste and packaging waste (Source: European Environment Agency¹⁵)

4. MUNICIPAL WASTE

In 2020, Europeans generated on average 521 kg of municipal waste per person, 49% of which was prepared for re-use or recycled while around 23% was landfilled¹⁶. The amount of waste generated has been slowly increasing in recent years.

Legal obligations on the management of municipal waste are laid down in the Waste Framework Directive. In advance of meeting the 55% target for the preparing for re-use and recycling of municipal waste¹⁷ to be achieved by 2025, Member States are required to meet a 50% target by 2020¹⁸. The Directive also includes longer-term ambitious targets: 60% by 2030 and 65% by 2035¹⁹, as well as other requirements such as implementing separate collection of waste and adopting national waste management plans and waste prevention programmes.

Based on the assessment carried out by the European Environment Agency and on an in-depth review of Member States' recycling performance and waste policies²⁰, **18 Member States** have been identified as **being at risk of missing the target of 55% preparing for re-use and recycling of municipal waste by 2025: Bulgaria, Croatia, Cyprus, Estonia, Finland, France, Greece, Hungary, Ireland, Latvia, Lithuania, Malta, Poland, Portugal, Romania, Slovakia, Spain and Sweden.**

Considering Member States' distance from the 2025 target and their specific challenges and situation, key country-specific **recommendations** to improve preparing for re-use and recycling rates have been identified. This has been done through a process that closely engaged the national authorities. If implemented swiftly by national and local authorities in cooperation with the waste management industry, these suggested measures could significantly accelerate the improvement of recycling performance. The staff working documents accompanying this report outline these possible actions for each Member State considered at risk. Key common priorities and related challenges are provided in the following box.

It may be reasonably assumed that this assessment also provides a strong indication of Member States' risk of missing the preparing for re-use and recycling target for 2030 of 60% if no additional efforts are made, including swifter implementation of adopted policies and reforms.

¹⁵ <https://www.eea.europa.eu/publications/many-eu-member-states-are>

¹⁶ Eurostat

¹⁷ Article 11(2)(a) of Directive 2008/98/EC.

¹⁸ Member States may choose among four methods to calculate the target for 2020, allowing to apply it to certain streams or to all municipal waste; the targets for 2025, 2030 and 2035 apply to all municipal waste.

¹⁹ Article 11(2)(c) to (e) of Directive 2008/98/EC.

²⁰ <https://www.eea.europa.eu/publications/many-eu-member-states-are>

Box 1: Municipal waste

- The majority of Member States have carried out or are in the process of implementing **waste reforms** that will result in a more **circular approach** and increased preparing for re-use and recycling rates in the years leading to 2025 and beyond. However, in some cases delays in their application are adversely affecting their performance. The challenges posed by the **COVID pandemic and the energy crisis** may require specific actions to avoid delayed implementation of national waste reforms.
- **Biowaste** is the most important waste stream for which action is needed, since it constitutes on average 34% of municipal waste. Focus should be given to introducing or **expanding effective capacity for the separate collection and treatment of biowaste**. Sound biowaste management could enable the use of biowaste as a fertiliser and soil improver and for the production of biogas²¹. It can provide significant contributions to **climate goals** and can help **achieve EU objectives for energy and critical raw materials security** by replacing mined mineral fertilisers while enhancing soil health.
- High capture rates and quality of separate collection are essential preconditions of preparing for re-use and recycling; this could be more effectively achieved via **targets for the mandatory separate collection of waste at municipal level**. Their use should be based on incentives and accountability (for instance, a system of financial rewards and penalties depending on the performance on those targets could be introduced).
- Measures are needed **to encourage households** to sort waste, including higher collection frequency for separated streams than for mixed waste.
- Member States are not using the full spectrum of **economic measures** to reduce landfilling and incineration (such as landfill and incineration taxes) and to enhance waste treatment associated with the higher steps of the waste hierarchy, such as preparing for re-use and recycling (for instance, pay-as-you-throw and deposit-return schemes).
- Member States need to close the investment gap including through the effective use of EU funds to develop **waste infrastructure that supports improving prevention, re-use and recycling performance**.
- Member States should take measures on the **prevention of non-recyclable waste**. Some Member States have not yet revised their national **waste prevention programmes**. The revision is expected to be completed by 2023.
- **Awareness-raising activities** are essential to enhance citizens' participation in better waste management. They can be tailored to different target groups (such as households, students, or tourists).

²¹ <https://www.eea.europa.eu/publications/biowaste-in-europe>

5. PACKAGING WASTE

In 2020, the total packaging waste generated in the EU amounted to around 79.6 million tonnes²². The average overall packaging recycling rate in the EU has steadily increased from 2005 to 2016 (67.6% in 2016) whereas a slight decrease has been observed from 2016 to 2020 (64% in 2020)²³. Furthermore, between 2013 and 2020 the amount of generated packaging waste grew by 15% across the EU, suggesting that more work on waste prevention is needed.

Legal obligations on the management of packaging waste are laid down in the Packaging Directive. These include a recycling target of 65% for all packaging waste to be achieved by 2025 and material-specific targets for paper and cardboard, glass, aluminium, plastic and wood. Member States are allowed to meet up to 5 percent of the targets by subtracting the share of reusable sales packaging placed on the market for the first time and re-used as part of a system to re-use packaging, and as regards wooden packaging, to take fully into account of repair for re-use. The Directive also includes longer-term ambitious recycling targets: 70% by 2030 for all packaging and also material-specific targets for paper and cardboard, glass, aluminium, plastic and wood²⁴.

Overall, according to the assessment carried out by the European Environment Agency²⁵, **10 Member States** have been identified as **at risk of missing the target of 65% recycling of packaging waste by 2025: Bulgaria, Croatia, Cyprus, Greece, Hungary, Lithuania, Malta, Poland, Romania, Slovakia.**

Furthermore, several Member States are considered to be at risk of missing one or more material-specific targets, with **plastic** being the most critical material:

Plastics	Austria, Bulgaria, Croatia, Cyprus, Denmark, Finland, France, Greece, Hungary, Ireland, Italy, Latvia, Luxembourg, Malta, Poland, Portugal, Romania, Slovakia and Spain
Glass	Bulgaria, Croatia, Cyprus, Greece, Hungary, Lithuania, Malta, Poland, Portugal and Romania
Aluminium	Croatia, Cyprus, Czechia, Greece, Malta, Portugal, Romania, Slovakia and Spain
Ferrous metal	Croatia, Denmark, Malta, Portugal and Romania
Paper and cardboard	Croatia, Malta, Slovakia and Spain
Wood	Croatia and Malta

The assessment identified some inconsistencies in packaging data reported by some Member States, indicating that the amounts of packaging put on the market may be underreported. In addition, the new recycling performance calculation requires Member States to apply as of 2020, as annual reporting of packaging waste generation and recycling is expected to result in

²² Eurostat

²³ Eurostat

²⁴ Article 6(1), points (h) and (i) of Directive 94/62/EC.

²⁵ <https://www.eea.europa.eu/publications/many-eu-member-states-are>

lower reported recycling rates by a few percentage points²⁶. At the same time, in case Member States take measures to ensure that re-use systems expand, they will have to meet lower recycling targets.

A preliminary review of data on packaging waste for reference year 2020 collected by Eurostat indicated that the assessment of Member States at risk of not meeting the targets remains valid. An overall slight reduction in the recycling rates can be observed (however, Member States showed very different patterns, with the performance of some increasing while that of others decreased). In addition, at this stage it is not yet possible to clearly attribute the changes to specific reasons due to several factors being at play, in particular the application of the new calculation method for recycling performance and the effects of the COVID pandemic.

The staff working documents accompanying this report outline possible actions that could be carried out to significantly reduce the risk of the waste packaging targets not being met. Some of the relevant common priorities identified in Box 1 above are also relevant for packaging. Specific challenges and priorities in relation to packaging are set out in Box 2 below.

²⁶ Directives (EU) 2018/850 and 2018/851 introduced new calculation rules requiring recycling to be measured at the point of packaging waste entering a recycling operation. The previous rules, reflected in the statistics up to 2019 and for some Member States for 2020, allow for all separately collected waste to be reported as recycled.

Box 2: Packaging waste

- **Continuous improvements in separate collection** remain essential to improve recycling performance. Separate collection improvements can be best achieved by a combination of a high share of the population being covered by high-convenience separate collection systems, the application of deposit-return schemes, the introduction of separate collection bins in public and commercial areas, and awareness raising.
- The majority of Member States struggle to meet the target on **plastic packaging**. This can be addressed by improving separate collection systems for plastic, and increasing plastic sorting and treatment capacities.
- Stepping up **efforts to establish re-use systems for packaging** will bring environmental benefits and help Member States in complying with the EU packaging recycling targets.
- Member States that perform weaker in terms of recycling often have **inconsistencies in packaging data**. This indicates that the amount of packaging put on the market may be underreported, with the result that statistics are inaccurate and that producers may not bear their full extended responsibility for financing waste collection, sorting and recycling.
- The impact of improvements in waste-management practices and infrastructure may be distorted in the reported data by the **application of the new calculation rules** applicable as of 2020. The impact of the new calculation rules cannot be fully assessed yet, but reported data by some Member States for 2020 indicate a decreasing effect of few percentage points.

6. LANDFILLING

Landfilling is the least preferable waste treatment option. The Landfill Directive²⁷ obliges Member States to reduce the landfilling of municipal waste to a maximum of 10% by 2035 and introduces a ban on the landfilling of separately collected waste, including biodegradable waste.

While amounts of landfilled municipal waste fell steadily in the EU as a whole until 2016 (dropping by 18% during the 2013-2016 period²⁸), the trend from 2017 to 2020 was not linear, with around 53.5 million tonnes of municipal waste landfilled in 2020. Furthermore, the average landfilling rate for municipal waste in the EU still stood at 23% in 2020. However, significant differences across the EU persist: in 2020 **eight Member States still landfilled over 50% of municipal waste**, with three of them reporting rates **above 70%**.

²⁷ Art. 5(5) of Directive 2018/850

²⁸ Eurostat.

A total of 13 Member States are still far from the target of a maximum of 10% of landfilling of municipal waste to be reached by 2035²⁹: **Bulgaria, Croatia, Cyprus, Czechia, Greece, Hungary, Latvia, Malta, Poland, Portugal, Romania, Slovakia and Spain.**

The distance to the target deserves further attention and requires the implementation of corrective measures, even if more time is needed to establish with more certainty the risk of missing the targets.

A study carried out for the Commission³⁰ also found that 15 Member States were not fully meeting the obligation laid down in the Directive³¹ to treat waste before landfilling.

Despite considerable improvements since the adoption of the Landfill Directive and closures of non-compliant landfills across the EU, the number of landfills that are **not in line with the requirements** of the Waste Framework and Landfill Directives remains a matter of concern. At the end of 2021, **1995 illegal or substandard landfills were still operating** and had to be rehabilitated or adapted to EU standards³².

7. BEST PRACTICES

Member States share common challenges in waste management and many solutions have been already found and applied. The wide sharing of such solutions is considered a priority. This report aims to identify a few best practices in municipal waste and packaging waste management from each Member State, providing an overview of successful initiatives and measures that could be replicated in other countries and could help Member States attain the targets.

The best practices collected address a wide range of subjects in waste management, such as governance, separate collection, including in remote areas, public-private partnerships, re-use and prevention of waste, waste treatment, communication and awareness raising, implementation of economic instruments, and effective extended producer responsibility schemes.

8. CONCLUSIONS

EU waste legislation has been driving considerable improvements in waste management, contributing to the EU circularity and zero pollution goals. However, full implementation of that legislation is crucial if the EU is to fully reap the environmental and economic benefits of a cleaner and more circular economy and compete in a world of increasingly scarce resources. Further efforts are also essential in the light of the substantial increase in packaging waste, the limited decrease in total waste generation (by 4% between 2010 and 2020), the pollution caused by marine plastic litter, and a relatively stable trend for residual

²⁹ <https://www.eea.europa.eu/publications/many-eu-member-states-are>

³⁰ Milieu (2017), 'Study to assess the implementation by the EU Member States of certain provisions of Directive 1999/31/EC on the landfill of waste'.

³¹ Article 6(a) of the Landfill Directive.

³² This figure is based on the number of illegal or substandard landfills which are subject to ongoing infringement procedures.

municipal waste, all of which put at risk the achievement of the zero pollution targets by 2030³³.

The Commission's early warning reports on waste and other compliance promotion activities point to continuous progress in Member States, albeit at different levels of performance. For the EU as a whole, the average rate of municipal waste preparing for re-use and recycling increased over 2010-2020, albeit slowly (from 37% to some 47%), while landfilling has significantly decreased, from 31% to some 23%. However, serious gaps, delays and challenges have also been identified and have to be swiftly addressed.

Good progress is possible if the Member States concerned take action to implement the measures identified in this report and in the accompanying country-specific staff working documents. Action must include a clear focus on biowaste and packaging waste. Improving waste management performance entails boosting separate collection, which is key to ensure the quality of waste and enable its recycling. In parallel, there is a pressing need to increase preparing for re-use and further develop waste treatment capacities for sorting and recycling in order to advance to the higher steps of the waste hierarchy.

Other important measures required concern the improvement of data quality, ensuring effective enforcement of the rules, and deploying enhanced economic instruments, such as efficient extended producer responsibility schemes and fit-for-purpose landfill and incineration taxes. These are all essential to fully reap the benefits of the circular economy and comply with EU waste legislation.

The Commission will continue to support Member States in their implementation efforts. This includes technical assistance, including through the Environmental Implementation Review³⁴, European Semester country recommendations³⁵, guidance on separate collection³⁶, exchange of best practices³⁷, reforms and investments under the Recovery and Resilience Facility, and financial support from other EU funds, such as the Cohesion Policy Funds (European Regional Development Fund and Cohesion Fund). Furthermore, the Commission has been launching other initiatives which contribute to a more circular economy and to less waste generated, such as the proposals for new regulations on packaging and packing waste and on ecodesign for sustainable products, and the revision of the Industrial Emissions Directive.

Lastly, recognising the wide differences in waste management situations, the EU rules on waste give some Member States the possibility to postpone compliance beyond the target years, under specific conditions. This option should be considered with care, since delays in the adoption of the necessary reforms will also delay environmental and economic benefits of the circular economy, including actions to reduce pressure on natural resources and dependencies on imports.

³³ https://environment.ec.europa.eu/strategy/zero-pollution-action-plan/zero-pollution-targets_en

³⁴ http://ec.europa.eu/environment/eir/index_en.htm

³⁵ https://commission.europa.eu/publications/2022-european-semester-country-specific-recommendations-commission-recommendations_en

³⁶ e.g. Guidance for separate collection of municipal waste <https://data.europa.eu/doi/10.2779/691513>

³⁷ TAIEX-EIR Peer2Peer - http://ec.europa.eu/environment/eir/p2p/index_en.htm